

DESIGN COM
froggy

July 10 (a) — © typeface by Chris Lozos

INSPIRED BY A TYPOPHILE BATTLE WEEK INVOLVING LIGATURE MADNESS WITH COOPER BLACK. FROGGY WAS MY DOUBLE “G” LIGATURE CARRIED ON TO SOMETHING ELSE, A SEMI-SERIF.

OHAMBURGEFONSTIV

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DEFGHIJKLMNOPQRSTUVWXYZ
AFTER WE LIVED AND REAPED
OUR BOUNTY, MANY YEARS
QUIETLY QUIETLY YIKES!**

Tačnije izražavanje istih. Naime naiđoh na jednu veoma interesantnu diskusiju na Typophile forumima o označavanju ironije u tekstu.

Koliko puta sam se samo osećao uskraćeno kada sam hteo misao da prenesem u pisanu formu, a na kraju ostao pitati se: “Da li će ovo neko uopšte da skapira?”. U svakom slučaju, radi se o arapskom znaku pitanja (koji je u suštini horizontalno izvrnut običan upitnik) koji bi se u engleskom (potencijalno i u drugim jezicima) koristio u svrhe označavanja ironije u tekstu.

Ovo je sada nameće pitanje da li je pametno uzimati karakter iz jednog jezika koji tamo služi jednoj svrsi, koji bi predstavljao nešto sasvim treće u nekom sasvim trećem jeziku? Da li to dovodi do dodatne konfuzije? Primer mogu biti navodnici recimo, u engleskom su oba navodnika gornja, a u srpskom, otvarajući navodnik ide dole, a zatvarajući gore. Nije li to komplikovano? Pogedajte samo Web kao primer—retko ko se bakće sa pravilnom tipografskom puntuacijom, a pogotovo se retko ko bavi pravilnom tipografskom puntuacijom u srpskom jeziku na Webu.

Ja se donekle trudim da ispoštujem tipografiju koliko mi vreme, poznavanje i sam CMS dozvoljavaju, i ujedno i apelujem na druge da pokušaju da urade isto. Gledam da primereno koristim duge i kratke srednje crte (engl. em i en dash), pravim razmak posle interpunkcijskih znakova, koristim elipsis za trotačku (da, trotačka nije tri obične tačke, već jedan karakter), pazim na dužinu pasusa, odvajam tekst u pasuse što je veoma bitno. Koliko puta sam samo dobio kilometarski elektronski email (da, i tako ga zovu) koji je napisan, što će reći u jednom dahu, tj. bez pasusa. Totalno se iznerviram kada mi se desi ovakva stvar. O pravopisu i da ne govorim.

Zanimljiv fenomen koji se nikako ne može zaobići jeste i opštepoznati i odveć omniprezentni smajli bez koga ne znam kakvo bi stanje i sporazumevanje bilo na webu.

magyar (Hungarian)

Károli Gáspár (1908)

Magyar Bibliatársulat újfordítású Bibliája (protestáns)

Szent István Társulati Biblia (katolikus)

Mind az egész földnek pedig egy nyelve és egyféle beszéde vala.

És lán mikor kelet felől elindultak vala, Sineár földén egy síkságot találának és ott letelepedének.

És mondának egymásnak: Jertek, vessünk téglát és égezzük ki jól; és lán nekik a téglakő gyanánt, a szurok pedig ragasztó gyanánt.

És mondának: Jertek, építsünk magunknak várost és tornyot, melynek teteje az eget érje, és szerezzünk magunknak nevet, hogy el ne széledjünk az egész földnek színén.

Az Úr pedig leszálla, hogy lássa a várost és a tornyot, melyet építenek vala az emberek fiai.

És monda az Úr: Imé e nép egy, s az egésznek egy a nyelve, és munkájának ez a kezdete; és bizony semmi sem gátolja, hogy véghez ne vigyenek mindent, a mit elgondolnak magukban.

Nosza szálljunk alá, és zavarjuk ott össze nyelvöket, hogy meg ne értsék egymás beszédét.

És elszéleszté őket onnan az Úr az egész földnek színére; és megszűnének építeni a várost.

Kekchi

Saß eb li cutan aßan jun ajcuiß li râtinobâleb li cristian saß ruchichßochß.

Li cristian queßcôeb chi cuânc saß li naßajej li cuan bar na-el cuiß chak li sakße. Queßxtau jun chßochß helho ru. Li chßochß aßan cuan xcuênt Sinar. Ut aran queßcana.

Ut queßxye chi ribileb rib: –Takayîb li xan ut takacßat chi us, chanqueb. Ut li xan quicßanjelac chokß rêkaj li pec. Ut li quicßanjelac re xletzbal li xan, aßan li na-el saß chßochß asfalto xcßababaß.

Queßxye: –Kayîbak junak katenamit ut jun li nimla ochoch najt xteram torre xcßababaß. Tâcuulak toj saß choxa li xnajtil xteram. Chi joßcaßin teßxqßue retal saß chixjunil li ruchichßochß nak lâo cuan kanaßleb. Ut incßaß takajeqßui kib yalak bar saß ruchichßochß, chanqueb.

Slovenský (Slovak)

Vtedy celý svet používal jedinú reč a rovnaké slová.

Keď sa na východe Ľudia pohli, našli rovinu v krajine Ľineár a usadili sa tam.

Povedali si: Poďme, narobme si tehál a dobre ich vypáďme. Tak im tehly slúžili za kameň a asfalt za maltu.

Potom si povedali: Poďme, vystavme si mesto a vežu, ktorej vrch by siahahal po nebesá; tak si urobíme meno, aby sme neboli roztratení po celej zemi.

Nato zostúpil Hospodin, aby videl mesto i vežu, ktorú stavali ľudskí synovia.

Hospodin riekol: Je tu jeden Ľud a všetci majú jednu reč; a to je len začiatok ich výčínov; teraz im už nič neznemožní vykonať, čo si zaumienia.

Poďme, zostúpme a zmäďme im tam reč, aby ani jeden nerozumel reči druhého.

Tak ich Hospodin rozptýlil po celej zemi a prestali stavať mesto.

Und sie sprachen einer zum anderen: Wohlan, laßt uns Ziegel streichen und hart brennen! Und der Ziegel diene ihnen als Stein, und das Erdharz diene ihnen als Mörtel.

Und sie sprachen: Wohlan, bauen wir uns eine Stadt und einen Turm, dessen Spitze an den Himmel reiche, und machen wir uns einen Namen, daß wir nicht zerstreut werden über die ganze Erde!

Und Jehova fuhr hernieder, die Stadt und den Turm zu sehen, welche die Menschenkinder bauten.

Und Jehova sprach: Siehe, sie sind ein Volk und haben alle eine Sprache, und dies haben sie angefangen zu tun; und nun wird ihnen nichts verwehrt werden, was sie zu tun ersinnen.

Wohlan, laßt uns herniederfahren und ihre Sprache daselbst verwirren, daß sie einer des anderen Sprache nicht verstehen!

Und Jehova zerstreute sie von dannen über die ganze Erde; und sie hörten auf, die Stadt zu bauen.

Darum gab man ihr den Namen Babel; (Verwirrung) denn daselbst verwirrte Jehova die Sprache der ganzen Erde, und von dannen zerstreute sie Jehova über die ganze Erde.

PRZETO ZGROMADZENIE OGŁALNE

Ogłasza Uroczyście niniejszą Powszechną Deklarację Praw Człowieka jako wspólny najwyższy cel wszystkich ludów i wszystkich narodów, aby wszyscy ludzie i wszystkie organy społeczeństwa – mając stale w pamięci niniejszą Deklarację – dążyły w drodze nauczania i wychowywania do rozwijania poszanowania tych praw i wolności i aby zapewniły za pomocą postępowych środków o zasięgu krajowym i międzynarodowym powszechnie i skuteczne uznanie i stosowanie tej Deklaracji zarówno wśród narodów Państw Członkowskich, jak i wśród narodów zamieszkujących obszary podległe ich władzy.

Artykuł 1

Wszyscy ludzie rodzą się wolni i równi pod względem swej godności i swych praw. Są oni obdarzeni rozumem i sumieniem i powinni postępować wobec innych w duchu braterstwa.

Każdy człowiek posiada wszystkie prawa i wolności zawarte w niniejszej Deklaracji bez względu na jakiejkolwiek różnice rasy, koloru, płci, języka, wyznania, poglądów politycznych i innych, narodowości, pochodzenia społecznego, majątku, urodzenia lub jakiegokolwiek innego stanu.

Nie wolno ponadto czynić żadnej różnicy w zależności od sytuacji politycznej, prawnej lub międzynarodowej kraju lub obszaru, do którego dana osoba przynależy, bez względu na to, czy dany kraj lub obszar jest niepodległy, czy też podlega systemowi powiernictwa, nie rządzi się samodzielnie lub jest w jakikolwiek sposób ograniczony w swej niepodległości.

FOCUS ON DARK MATTER

KLAUS PRETZL 2000 NEW J. PHYS.
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THIS 'FOCUS ON DARK MATTER' CLUSTER OF ARTICLES IS THE VERY FIRST IN A SERIES OF FEATURED TOPICS TO APPEAR IN NEW JOURNAL OF PHYSICS THAT WILL BE OF GREAT INTEREST TO THE PHYSICS COMMUNITY. THE IDEA IS TO BRING TOGETHER CONTRIBUTIONS FROM LEADING RESEARCHERS IN TOPICAL FIELDS TO PROVIDE INSIGHTS INTO THE KEY ISSUES FOR BOTH EXPERTS AND NON-SPECIALISTS ALIKE.

INVESTIGATIONS OF DARK MATTER IN THE UNIVERSE ARE OF FUNDAMENTAL INTEREST TO ASTRONOMERS, ASTROPHYSICISTS, COSMOLOGISTS, AND NUCLEAR AND PARTICLE PHYSICISTS. RECENT DEVELOPMENTS HAVE PROVIDED NEW INSIGHT INTO THE MATTER BUDGET OF THE UNIVERSE. USING SUPERNOVAE OF TYPE 1A AS STANDARD CANDLES, DISTANCE MEASUREMENTS INDICATE AN ACCELERATING EXPANSION OF THE UNIVERSE. THIS RESULT SUGGESTS A NON-ZERO COSMOLOGICAL CONSTANT, AND THE RESULTING VACUUM ENERGY DENSITY CONTRIBUTES 70% OF THE CRITICAL MASS OF THE UNIVERSE. GRAVITATIONAL LENSING, ORIGINALLY PROPOSED BY EINSTEIN AND LATER UTILIZED FOR

MASS DETERMINATION OF GALACTIC CLUSTERS BY F ZWICKY AND OTHERS, HAS RECENTLY BEEN USED TO MAP DARK MATTER IN CLUSTERS AND LARGE-SCALE STRUCTURES. THESE EXPERIMENTS SHOW THAT 30% OF THE CRITICAL MASS IS CONTAINED IN THESE STRUCTURES, OF WHICH ONLY 3% IS VISIBLE. SINCE THE BARYONIC COMPONENT OF MATTER IS ONLY 6% OF THE CRITICAL MASS, ASSUMING A HUBBLE CONSTANT OF $60 \text{ km s}^{-1} \text{ Mpc}^{-1}$, THERE IS FIVE TIMES MORE DARK MATTER THAN BARYONIC MATTER IN THE UNIVERSE. PRIME CANDIDATES FOR MAKING UP THIS DEFICIT ARE NEUTRINOS WITH MASS (HOT DARK MATTER) AND WEAKLY INTERACTING MASSIVE PARTICLES, THE SO-CALLED WIMPS (COLD DARK MATTER). COSMOLOGISTS AND PARTICLE PHYSICISTS FAVOUR THE LATTER TO BE THE LIGHTEST SUPERSYMMETRIC PARTICLE, THE SO-CALLED NEUTRALINO, SINCE IF IT DOES EXIST IT WOULD SOLVE TWO PROBLEMS AT THE SAME TIME, NAMELY COLD DARK MATTER AND GRAND UNIFICATION THEORIES. COSMIC BACKGROUND MICROWAVE RADIATION DATA AND LARGE-SCALE STRUCTURE DISTRIBUTIONS, HOWEVER, FAVOUR A COCKTAIL OF 30% HOT AND 70% COLD DARK MATTER.

TREUMANN'S GROUP INVESTIGATES MATTER DISTRIBUTIONS IN GALACTIC CLUSTERS, INCLUDING MASSIVE NEUTRINOS AND COLLISIONAL INTRACLUSTER GAS EMITTING X-RAY RADIATION. THE ISSUE OF BARYONIC DARK MATTER IN OUR GALACTIC HALO IS EXPLORED IN THE CONTRIBUTION FROM DE PAOLIS AND COWORKERS. THEY SHOW THAT THE OBSERVED DIFFUSE GAMMA-RAY EMISSION FROM THE GALACTIC HALO CAN BE EXPLAINED BY COSMIC RAY PROTON SCATTERING ON CLOUDS OF DARK CLUSTERS OF BROWN DWARFS AND COLD SELF-GRAVITATING H_2 CLOUDS IN THE HALO. NEW EXPERIMENTAL RESULTS ON TERRESTRIAL WIMP SEARCHES VIA WIMP-XE INELASTIC SCATTERING ARE PRESENTED BY BERNABEI'S GROUP. THE CONTRIBUTION FROM CEBRIÁN'S TEAM DESCRIBES PRESENT AND FUTURE WIMP EXPERIMENTS USING CONVENTIONAL AND CRYOGENIC DETECTORS IN THE CANFRANC UNDERGROUND LABORATORY. AN INTERESTING NOVEL LARGE-MASS, LOW-BACKGROUND, SUPERHEATED DROPLET DETECTOR FOR WIMP SEARCHES IS DESCRIBED BY COLLAR AND COLLEAGUES.

PHYSICS GETS DARK AND EXOTIC

EDITORIAL: JANUARY 2000

TWO OF THE OUTSTANDING CHALLENGES IN PHYSICS IDENTIFIED IN OUR MILLENNIUM SURVEY LAST MONTH WERE THE NATURE OF “DARK MATTER” AND A PROPER UNDERSTANDING OF NUCLEAR STRUCTURE. THIS MONTH WE LOOK AT THESE CHALLENGES IN GREATER DETAIL.

DARK MATTER IS MATTER THAT DOES NOT INTERACT WITH ELECTROMAGNETIC RADIATION: IT CANNOT BE SEEN WITH TELESCOPES AND ONLY REVEALS ITSELF THROUGH ITS GRAVITATIONAL INTERACTIONS. ASTRONOMERS FIRST BECAME AWARE IN THE 1930S THAT OUR MILKY WAY GALAXY WAS ROTATING FASTER THAN COULD BE EXPLAINED BY THE GRAVITATIONAL INFLUENCE OF THE STARS AND DUST THAT IT CONTAINED. IT LATER BECAME CLEAR THAT MORE THAN 90% OF THE MATTER IN THE UNIVERSE—AND POSSIBLY AS MUCH AS 99%—MIGHT BE DARK.

SOME OF THE DARK MATTER MIGHT BE IN THE FORM OF ORDINARY OR “BARYONIC” MATTER MADE PRIMARILY OF NEUTRONS AND PROTONS—SUCH AS FAILED STARS OR BLACK HOLES. A SMALL FRACTION MIGHT BE IN THE FORM OF NEUTRINOS, BUT THE MAJORITY OF DARK MATTER IS NON-BARYONIC AND BEYOND THE STANDARD MODEL OF PARTICLE PHYSICS. NEUTRINOS WITH MASS ARE ALSO BEYOND THE STANDARD MODEL BUT NOT AS FAR BEYOND AS NEUTRALINOS, AXIONS AND OTHER EXOTIC PARTICLES. ANOTHER POSSIBILITY FOR THIS MISSING MASS IS “DARK ENERGY” IN THE FORM OF A COSMOLOGICAL CONSTANT.

IN THIS ISSUE NIGEL SMITH AND NEIL SPOONER OF THE UK DARK MATTER COLLABORATION DESCRIBE EFFORTS TO DETECT DARK MATTER IN UNDERGROUND EXPERIMENTS. ALTHOUGH THE EXPERIMENTS SOUND EASY—CONNECT SOME PHOTOMULTIPLIER TUBES TO A CRYSTAL AND COUNT THE FLASHES—THE CHALLENGE OF ISOLATING A DARK-MATTER SIGNAL AMONGST A MYRIAD OF OTHER BACKGROUND EFFECTS IS FORMIDABLE. HOWEVER, DARK-MATTER SEARCHES CAN BE PERFORMED BY TEAMS MUCH SMALLER THAN THOSE FOUND IN TRADITIONAL PARTICLE-PHYSICS EXPERIMENTS. THE UNEQUIVOCAL DETECTION OF A DARK-MATTER PARTICLE WOULD BE A TREMENDOUS BOOST FOR PARTICLE PHYSICS AND COSMOLOGY, AL-

THOUGH THERE WOULD DOUBTLESS BE CALLS FOR IT TO BE CONFIRMED IN AN ACCELERATOR-BASED EXPERIMENT.

BACK IN THE BARYONIC WORLD, MEANWHILE, NUCLEAR PHYSICISTS ARE PROBING MORE AND MORE EXOTIC NUCLEI. OVER 7000 DIFFERENT NUCLEI ARE THOUGHT TO EXIST, BUT ONLY 3000 OF THESE HAVE BEEN CREATED AND STUDIED IN THE LABORATORY, AND ONLY 260 OF THE TOTAL ARE STABLE. AS ISAO TANIHATA EXPLAINED LAST MONTH, THERE ARE VARIOUS MODELS OF THE NUCLEUS THAT WORK WELL IN DIFFERENT REGIONS OF THE CHART OF THE NUCLIDES—SOMETHING THAT IS A SOURCE OF BOTH PRIDE AND FRUSTRATION TO NUCLEAR PHYSICISTS. IN THIS ISSUE PADDY REGAN AND BERTRAM BLANK DESCRIBE ATTEMPTS TO SYNTHESIZE AND STUDY NUCLEI AT THE VERY LIMITS OF STABILITY—THE PROTON AND NEUTRON DRIPLINES.

THESE EXOTIC SPECIES CAN BE DETECTED AMONG THE DEBRIS OF COLLISIONS BETWEEN STABLE NUCLEI. HOWEVER, IF ENOUGH OF THESE SHORT-LIVED NUCLEI COULD BE ISOLATED AND THEN RE-ACCELERATED IN A RADIOACTIVE BEAM, IT WOULD BE POSSIBLE TO CREATE AND STUDY EVEN MORE EXOTIC NUCLEI. PLANS FOR NEW FACILITIES TO PERFORM SUCH EXPERIMENTS ARE WELL ADVANCED IN VARIOUS NUCLEAR-PHYSICS LABS AROUND THE WORLD.

AS WE EMBARK ON THE 21ST CENTURY, DARK MATTER AND NUCLEAR STRUCTURE COULD BE TWO OF THE OUTSTANDING CHALLENGES IN PHYSICS TO BE OVERCOME FIRST.

ULTIMATE LIMITS TO INERTIAL MASS SENSING BASED UPON NANOELECTROMECHANICAL SYSTEMS

AUTHORS: EKINCI, K. L.; YANG, Y. T.; ROUKES, M. L.

ABSTRACT

NANOMECHANICAL RESONATORS CAN NOW BE REALIZED THAT ACHIEVE FUNDAMENTAL RESONANCE FREQUENCIES EXCEEDING 1 GHz, WITH QUALITY FACTORS (Q) IN THE RANGE $10^3 \leq Q \leq 10^5$. THE MINUSCULE ACTIVE MASSES OF THESE DEVICES, IN CONJUNCTION WITH THEIR HIGH Q S, TRANSLATE INTO UNPRECEDENTED INERTIAL MASS SENSITIVITIES. THIS MAKES THEM NATURAL CANDIDATES FOR A VARIETY OF MASS SENSING APPLICATIONS. HERE WE EVALUATE THE ULTIMATE MASS SENSITIVITY LIMITS FOR NANOMECHANICAL RESONATORS OPERATING IN VACUO THAT ARE IMPOSED BY A NUMBER OF FUNDAMENTAL PHYSICAL NOISE PROCESSES. OUR ANALYSES INDICATE THAT NANOMECHANICAL RESONATORS OFFER IMMENSE POTENTIAL FOR MASS SENSING-ULTIMATELY WITH RESOLUTION AT THE LEVEL OF INDIVIDUAL MOLECULES.

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